

Take-Home Assignment

AI Engineer Interview — Internal Corporate Chatbot

3–5 days

Senior /
Mid-level

CONTEXT

Your task is to design and prototype an internal corporate chatbot that allows employees to query internal knowledge bases (HR policies, technical documentation, project wikis, etc.). You are not expected to build a fully production-ready system — we want to evaluate your thinking, architectural judgment, and hands-on AI engineering skills.

DELIVERABLES



1 — Architecture overview

Produce a written or diagrammatic overview of your system design. Must cover:

- End-to-end data flow from user query to response
- Ingestion pipeline for internal documents (crawling, chunking, embedding)
- Retrieval strategy (vector search, hybrid, re-ranking, etc.)
- LLM orchestration layer (prompt construction, context management)
- Authentication and access-control model for sensitive docs
- How you would handle multi-tenancy or department-level isolation

2 — Tech stack selection

Document each layer of the stack with clear rationale. Address:

- LLM provider and model choice (with justification vs. alternatives)
- Embedding model and vector database
- Orchestration framework (e.g. LangChain, LlamaIndex, custom)
- Document parsing and chunking strategy
- Backend API, hosting / infrastructure
- Frontend or interface layer (even if minimal)
- Observability, logging, and eval tooling

■ *For each major choice, briefly explain what you considered and why you chose it over the alternative.*

3 — Implementation approach

Describe how you would build this in practice. Include:

- A working prototype or proof-of-concept (even on toy data)
- RAG pipeline: chunking strategy, embedding, retrieval, prompt design
- How you handle ambiguous or out-of-scope queries
- Evaluation approach — how do you measure answer quality?
- Key risks and open questions you'd investigate in a real engagement
- What you'd prioritize in a 4-week vs. 3-month build

4 — Cost estimate

Provide a realistic cost model for a mid-sized company (500–2,000 employees). Break down:

- One-time build costs — engineering hours, infra setup, indexing
- Ongoing monthly costs — LLM tokens, vector DB storage, hosting
- Cost levers — caching, model tiering, query volume
- Assumptions — state them clearly and show your math

■ *We don't expect exact figures. We're evaluating whether you understand the cost drivers and can reason about tradeoffs.*

5 — Submission format

Package your work as:

- A public GitHub repo (or zip) with a README.md as the entry point
- Architecture diagram (any format: Mermaid, draw.io, Excalidraw, image)
- Working prototype code, even if partial or using mock data
- Cost estimate in a doc, spreadsheet, or README section
- A short 5–10 min Loom walkthrough is encouraged but not required

BONUS CONSIDERATIONS (NOTICED, NOT SCORED)

Streaming responses	Citation / source attribution	Conversation memory	Feedback / thumbs-down loop
Role-based doc access	Hallucination mitigation	Latency benchmarks	Offline / on-prem variant considered

WHAT WE'RE NOT LOOKING FOR

A perfect, fully deployed system. We want to see how you think, what you prioritize, where you cut scope intentionally, and how clearly you communicate tradeoffs. A thoughtful incomplete design beats a shallow but "complete" one.